CM0845 Logic
Propositional Logic: Natural Deduction

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Preliminaries

Textbook

D. van Dalen (2013). Logic and Structure. Sections 2.4 and 2.6.

Convention

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

Derivation Rules for $\{\wedge, \to, \perp\}$

Conjunction

$$\frac{\varphi \qquad \psi}{\varphi \wedge \psi}\,\wedge\mathrm{I} \qquad\qquad \frac{\varphi \wedge \psi}{\varphi}\,\wedge\mathrm{E} \qquad\qquad \frac{\varphi \wedge \psi}{\psi}\,\wedge\mathrm{E}$$

Derivation Rules for $\{\wedge, \rightarrow, \perp\}$

Implication

$$\dfrac{\begin{array}{c}[\varphi]^x \\ \vdots \\ \psi\end{array}}{\varphi \rightarrow \psi} \rightarrow\mathrm{I}^x \qquad\qquad \dfrac{\varphi \qquad \varphi \rightarrow \psi}{\psi} \rightarrow\mathrm{E}$$

Remark: In the application of the $\rightarrow\mathrm{I}$ rule, we may discharge zero, one, or more occurrences of the assumption.

Derivation Rules for $\{\wedge, \to, \perp\}$

Bottom elimination

$$\frac{\perp}{\varphi}\ \perp\mathrm{E}$$

Proof by contradiction (*reductio ad absurdum*)

$$\begin{array}{c} [\neg\varphi]^x \\ \vdots \\ \dfrac{\perp}{\varphi}\ \mathrm{RAA}^x \end{array}$$

where $\neg\varphi := \varphi \to \perp$.

Derivation Rules for $\{\wedge, \rightarrow, \perp\}$

Definition 2.4.2

Let Γ be a set of formulae and let φ be a formula. The relation $\Gamma \vdash \varphi$ means that there is a derivation with conclusion φ from the set of hypotheses Γ.

Example

A derivation where every assumption is discharged once. A proof of Pierce's law $\vdash ((\varphi \rightarrow \psi) \rightarrow \varphi) \rightarrow \varphi$.[†]

Proof

$$
\begin{prooftree}
\AxiomC{$[\varphi]^x$}
\AxiomC{$[\neg\varphi]^y$}
\RightLabel{$\rightarrow$E}
\BinaryInfC{$\bot$}
\RightLabel{$\bot$E}
\UnaryInfC{ψ}
\RightLabel{$\rightarrow$I^x}
\UnaryInfC{$\varphi \rightarrow \psi$}
\AxiomC{$[(\varphi \rightarrow \psi) \rightarrow \varphi]^z$}
\RightLabel{$\rightarrow$E}
\BinaryInfC{φ}
\AxiomC{$[\neg\varphi]^y$}
\RightLabel{$\rightarrow$E}
\BinaryInfC{$\bot$}
\RightLabel{RAA^y}
\UnaryInfC{φ}
\RightLabel{$\rightarrow$I^z}
\UnaryInfC{$(\varphi \rightarrow \psi) \rightarrow \varphi) \rightarrow \varphi$}
\end{prooftree}
$$

[†] Adapted from (Carr 2018).

Derivation Rules for $\{\wedge, \rightarrow, \bot\}$

Example

A derivation using the same assumption twice. A proof that $\vdash (\varphi \wedge \psi) \rightarrow (\psi \wedge \varphi)$.

Proof

$$\dfrac{\dfrac{\dfrac{[\varphi \wedge \psi]^x}{\psi}\,\wedge\mathrm{E} \qquad \dfrac{[\varphi \wedge \psi]^x}{\varphi}\,\wedge\mathrm{E}}{\psi \wedge \varphi}\,\wedge\mathrm{I}}{(\varphi \wedge \psi) \rightarrow (\psi \wedge \varphi)}\,\rightarrow\mathrm{I}^x$$

Derivation Rules for $\{\wedge, \to, \perp\}$

Example

A derivation where the assumption and the conclusion are the same. A proof that $\vdash \varphi \to \varphi$.

Proof

$$\cfrac{[\varphi]^x}{\varphi \to \varphi}\, {\to}\mathrm{I}^x$$

Derivation Rules for $\{\wedge, \rightarrow, \perp\}$

Remark

"The rule schemes of natural deduction display only the open assumptions that are **active** in the rule, but there may be any number of other assumptions." (Negri and von Plato 2008, p. 10)

Derivation Rules for $\{\wedge, \rightarrow, \bot\}$

Remark

"The rule schemes of natural deduction display only the open assumptions that are **active** in the rule, but there may be any number of other assumptions." (Negri and von Plato 2008, p. 10)

Example

A derivation where there is a vacuous discharge when using the inference rule $\rightarrow\mathrm{I}$. A proof that $\vdash \varphi \rightarrow (\psi \rightarrow \varphi)$.

Proof

$$\cfrac{\cfrac{[\varphi]^x}{\psi \rightarrow \varphi}\ \rightarrow\mathrm{I}\ \text{(vacuous discharge of } \psi\text{)}}{\varphi \rightarrow (\psi \rightarrow \varphi)}\ \rightarrow\mathrm{I}^x$$

Derivation Rules for $\{\wedge, \rightarrow, \bot\}$

Example

A derivation using one hypothesis. A proof that $\varphi \vdash \neg(\neg\varphi \wedge \psi)$ (van Dalen 2013, Exercise 3.(a), p. 37).

Proof

$$\dfrac{\varphi \qquad \dfrac{[\neg\varphi \wedge \psi]^x}{\neg\varphi}\,\wedge\mathrm{E}}{\dfrac{\bot}{\neg(\neg\varphi \wedge \psi)}\,{\rightarrow}\mathrm{I}^x}\,{\rightarrow}\mathrm{E}$$

Derivation Rules for $\{\wedge, \to, \bot\}$

Example

A derivation using the same hypothesis twice. A proof that $\varphi \wedge \psi \vdash \psi \wedge \varphi$.

Proof

$$\dfrac{\dfrac{\varphi \wedge \psi}{\psi}\,\wedge\mathrm{E} \qquad \dfrac{\varphi \wedge \psi}{\varphi}\,\wedge\mathrm{E}}{\psi \wedge \varphi}\,\wedge\mathrm{I}$$

Set of Derivations

Notation

(Whiteboard)

Definition 2.4.1

The **set of derivations**, denoted $\mathcal{D}$, is the smallest set X with the properties:

(see next slide)

Set of Derivations

(1) The one-element tree φ belongs to X for all $\varphi \in PROP$.

(2$\wedge$) If $\begin{array}{c}\mathcal{D}\\ \varphi\end{array}, \begin{array}{c}\mathcal{D}'\\ \varphi'\end{array} \in X$, then $\dfrac{\begin{array}{cc}\mathcal{D} & \mathcal{D}'\\ \varphi & \varphi'\end{array}}{\varphi\wedge\varphi'} \in X$.

If $\begin{array}{c}\mathcal{D}\\ \varphi\wedge\psi\end{array} \in X$, then $\dfrac{\begin{array}{c}\mathcal{D}\\ \varphi\wedge\psi\end{array}}{\varphi}, \dfrac{\begin{array}{c}\mathcal{D}\\ \varphi\wedge\psi\end{array}}{\psi} \in X$.

(2$\rightarrow$) If $\begin{array}{c}\varphi\\ \mathcal{D}\\ \psi\end{array} \in X$, then $\dfrac{\begin{array}{c}[\varphi]\\ \mathcal{D}\\ \psi\end{array}}{\varphi\rightarrow\psi} \in X$.

If $\begin{array}{c}\mathcal{D}\\ \varphi\end{array}, \begin{array}{c}\mathcal{D}'\\ \varphi\rightarrow\psi\end{array} \in X$, then $\dfrac{\begin{array}{cc}\mathcal{D} & \mathcal{D}'\\ \varphi & \varphi\rightarrow\psi\end{array}}{\psi} \in X$.

(2$\bot$) If $\begin{array}{c}\mathcal{D}\\ \bot\end{array} \in X$, then $\dfrac{\begin{array}{c}\mathcal{D}\\ \bot\end{array}}{\varphi} \in X$.

If $\begin{array}{c}\neg\varphi\\ \mathcal{D}\\ \bot\end{array} \in X$, then $\underline{\begin{array}{c}[\neg\varphi]\\ \mathcal{D}\\ \bot\end{array}} \in X$.

Derivation Rules for the Missing Connectives $\{\vee, \neg, \leftrightarrow\}$

Disjunction

$$\frac{\varphi}{\varphi \vee \psi}\ \vee\mathrm{I} \qquad \frac{\psi}{\varphi \vee \psi}\ \vee\mathrm{I} \qquad \frac{\varphi \vee \psi \quad \begin{matrix}[\varphi]^x \\ \vdots \\ \sigma\end{matrix} \quad \begin{matrix}[\psi]^y \\ \vdots \\ \sigma\end{matrix}}{\sigma}\ \vee\mathrm{E}^{x,y}$$

Derivation Rules for the Missing Connectives $\{\vee, \neg, \leftrightarrow\}$

Negation

$$\dfrac{\begin{matrix}[\varphi]^x \\ \vdots \\ \bot\end{matrix}}{\neg\varphi}\ \neg\mathrm{I}^x \qquad\qquad \dfrac{\varphi \qquad \neg\varphi}{\bot}\ \neg\mathrm{E}$$

Derivation Rules for the Missing Connectives $\{\vee, \neg, \leftrightarrow\}$

Equivalence

$$\dfrac{\begin{matrix}[\varphi]^x \\ \vdots \\ \psi\end{matrix} \qquad \begin{matrix}[\psi]^y \\ \vdots \\ \varphi\end{matrix}}{\varphi \leftrightarrow \psi} \leftrightarrow\mathrm{I}^{\mathrm{x,y}} \qquad \dfrac{\varphi \qquad \varphi \leftrightarrow \psi}{\psi} \leftrightarrow\mathrm{E} \qquad \dfrac{\psi \qquad \varphi \leftrightarrow \psi}{\varphi} \leftrightarrow\mathrm{E}$$

Derivation Rules for $\{\wedge, \vee, \to, \bot\}$

$$\dfrac{\varphi \qquad \psi}{\varphi \wedge \psi}\;\wedge\mathrm{I}$$

$$\dfrac{\varphi \wedge \psi}{\varphi}\;\wedge\mathrm{E} \qquad \dfrac{\varphi \wedge \psi}{\psi}\;\wedge\mathrm{E}$$

$$\dfrac{\varphi}{\varphi \vee \psi}\;\vee\mathrm{I} \qquad \dfrac{\psi}{\varphi \vee \psi}\;\vee\mathrm{I}$$

$$\dfrac{\varphi \vee \psi \qquad \begin{matrix}[\varphi]^x \\ \vdots \\ \sigma\end{matrix} \qquad \begin{matrix}[\psi]^y \\ \vdots \\ \sigma\end{matrix}}{\sigma}\;\vee\mathrm{E}^{x,y}$$

$$\dfrac{\begin{matrix}[\varphi]^x \\ \vdots \\ \psi\end{matrix}}{\varphi \to \psi}\;\to\mathrm{I}^x$$

$$\dfrac{\varphi \qquad \varphi \to \psi}{\psi}\;\to\mathrm{E}$$

$$\dfrac{\bot}{\varphi}\;\bot\mathrm{E}$$

$$\begin{matrix}[\neg\varphi]^x \\ \vdots\end{matrix}$$

Example

Prove that $\vdash \varphi \vee \neg\varphi$ (van Dalen 2013, example p. 49).

Proof

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{[\varphi]^x}{\varphi \vee \neg\varphi}\,\vee\mathrm{I} \qquad [\neg(\varphi \vee \neg\varphi)]^y}{\bot}\,{\to}\mathrm{E}}{\neg\varphi}\,{\to}\mathrm{I}^x}{\varphi \vee \neg\varphi}\,\vee\mathrm{I} \qquad [\neg(\varphi \vee \neg\varphi)]^y}{\bot}\,{\to}\mathrm{E}}{\varphi \vee \neg\varphi}\,\mathrm{RAA}^y$$

Natural Deduction in Sequent Calculus Style

$$\frac{}{\Gamma, \varphi \vdash \varphi}\ \mathrm{Ax}$$

$$\frac{\Gamma \vdash \varphi \qquad \Gamma \vdash \psi}{\Gamma \vdash \varphi \wedge \psi}\ \wedge\mathrm{I} \qquad \frac{\Gamma \vdash \varphi \wedge \psi}{\Gamma \vdash \varphi}\ \wedge\mathrm{E} \qquad \frac{\Gamma \vdash \varphi \wedge \psi}{\Gamma \vdash \psi}\ \wedge\mathrm{E}$$

$$\frac{\Gamma \vdash \varphi}{\Gamma \vdash \varphi \vee \psi}\ \vee\mathrm{I} \qquad \frac{\Gamma \vdash \psi}{\Gamma \vdash \varphi \vee \psi}\ \vee\mathrm{I} \qquad \frac{\Gamma \vdash \varphi \vee \psi \quad \Gamma, \varphi \vdash \sigma \quad \Gamma, \psi \vdash \sigma}{\Gamma \vdash \sigma}\ \vee\mathrm{E}$$

$$\frac{\Gamma, \varphi \vdash \psi}{\Gamma \vdash \varphi \rightarrow \psi}\ {\rightarrow}\mathrm{I} \qquad \frac{\Gamma \vdash \varphi \qquad \Gamma \vdash \varphi \rightarrow \psi}{\Gamma \vdash \psi}\ {\rightarrow}\mathrm{E}$$

$$\frac{\Gamma \vdash \bot}{\Gamma \vdash \varphi}\ \bot\mathrm{E} \qquad \frac{\Gamma, \neg\varphi \vdash \bot}{\Gamma \vdash \varphi}\ \mathrm{RAA}$$

Natural Deduction in Sequent Calculus Style

Example

We prove that $\vdash \varphi \vee \neg\varphi$.

Proof

Let $\Gamma = \{\varphi, \neg(\varphi \vee \neg\varphi)\}$ and $\Delta = \Gamma - \{\varphi\}$.

$$
\begin{prooftree}
\AxiomC{}
\RightLabel{Ax}
\UnaryInfC{$\Gamma \vdash \varphi$}
\RightLabel{$\vee$I}
\UnaryInfC{$\Gamma \vdash \varphi \vee \neg\varphi$}
\AxiomC{}
\RightLabel{Ax}
\UnaryInfC{$\Gamma \vdash \neg(\varphi \vee \neg\varphi)$}
\RightLabel{$\to$E}
\BinaryInfC{$\Gamma \vdash \bot$}
\RightLabel{$\to$I}
\UnaryInfC{$\Delta \vdash \neg\varphi$}
\RightLabel{$\vee$I}
\UnaryInfC{$\Delta \vdash \varphi \vee \neg\varphi$}
\AxiomC{}
\RightLabel{Ax}
\UnaryInfC{$\Delta \vdash \neg(\varphi \vee \neg\varphi)$}
\RightLabel{$\to$E}
\BinaryInfC{$\Delta \vdash \bot$}
\end{prooftree}
$$

Example

A derivation where there is a vacuous discharge when using the inference rule $\to\mathrm{I}$. A proof that $\vdash \varphi \to (\psi \to \varphi)$.

Proof

$$\dfrac{\dfrac{\dfrac{}{\varphi \vdash \varphi}\,\mathrm{Ax}}{\varphi \vdash \psi \to \varphi}\,\to\mathrm{I}\ \text{(vacuous discharge of } \psi\text{)}}{\vdash \varphi \to (\psi \to \varphi)}\,\to\mathrm{I}$$

References

Alastair Carr (Mar. 2018). Natural Deduction Pack. URL: https://github.com/Alastair-Carr/Natural-Deduction-Pack (visited on 06/08/2025) (cit. on p. 7).

Sara Negri and Jan von Plato [2001] (2008). Structural Proof Theory. Digitally printed version. Cambridge University Press. DOI: 10.1017/CBO9780511527340 (cit. on pp. 10, 11).

Dirk van Dalen [1980] (2013). Logic and Structure. 5th ed. Springer (cit. on pp. 2, 12, 20).